

**Department of Computer Engineering COMSATS University
Islamabad, Attock Campus**

Lab Rubrics Form	
Lab Name	Discrete-Time Signal Processing — Differentiator and Accumulator Implementation
Lab Number	08
Submitted by:	
Student Names	Registration #
Sumaira Safeer	FA22-BCE-019

Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab-08

Discrete-Time Signal Processing — Differentiator and Accumulator Implementation

Objective:

The objective of this lab is to understand and implement two fundamental discrete-time signal processing systems:

1. **Differentiator:** To compute the first-order difference of a discrete-time signal, defined as:

$$y[n] = x[n] - x[n-1]$$

This operation approximates the derivative and highlights changes in the signal.

2. **Accumulator:** To compute the running sum (discrete-time integration) of a discrete-time signal, defined as:

$$z[n] = \sum_{k=-\infty}^n x[k]$$

This operation accumulates the signal values over time, analogous to integration in continuous-time systems.

The lab also aims to verify the inverse relationship between the differentiator and accumulator systems through implementation and analysis of a given triangular signal $x[n] = |n|$ for $-3 \leq n \leq 3$.

Lab Task:

Consider the discrete-time signal and implement a differentiator and accumulator.

Original Signal

$$x[n] = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Code:

```
clear; close all; clc;
% Define range
n = -10:10;
% Original signal x[n] = |n| for -3 ≤ n ≤ 3
x = zeros(size(n));
for k = 1:length(n)
    if n(k) >= -3 && n(k) <= 3
```

```

        x(k) = abs(n(k));
    end
end
% Differentiator:  $y[n] = x[n] - x[n-1]$ 
y = zeros(size(n));
for k = 2:length(n)
    y(k) = x(k) - x(k-1);
end
% Handle boundary (n = -10)
y(1) = x(1);
% Accumulator:  $z[n] = \sum_{k=-\infty}^n x[k]$ 
z = zeros(size(n));
cum_sum = 0;
for k = 1:length(n)
    cum_sum = cum_sum + x(k);
    z(k) = cum_sum;
end
% Display values for verification
fprintf('n\t x[n]\t y[n]\t z[n]\n');
for idx = find(n >= -4 & n <= 4)
    fprintf('%d\t %d\t %d\t %d\n', n(idx), x(idx), y(idx), z(idx));
end
% Plot all signals
figure('Position', [100, 100, 800, 900]);
% Plot x[n]
subplot(3,1,1);
stem(n, x, 'b', 'filled', 'LineWidth', 2);
title('x[n] = |n| for -3 ≤ n ≤ 3', 'FontSize', 12);
xlabel('n'); ylabel('Amplitude');
grid on;
ylim([0 4]);
xlim([-10 10]);

% Plot y[n] (differentiator)
subplot(3,1,2);
stem(n, y, 'r', 'filled', 'LineWidth', 2);
title('y[n] = x[n] - x[n-1] (Differentiator)', 'FontSize', 12);
xlabel('n'); ylabel('Amplitude');

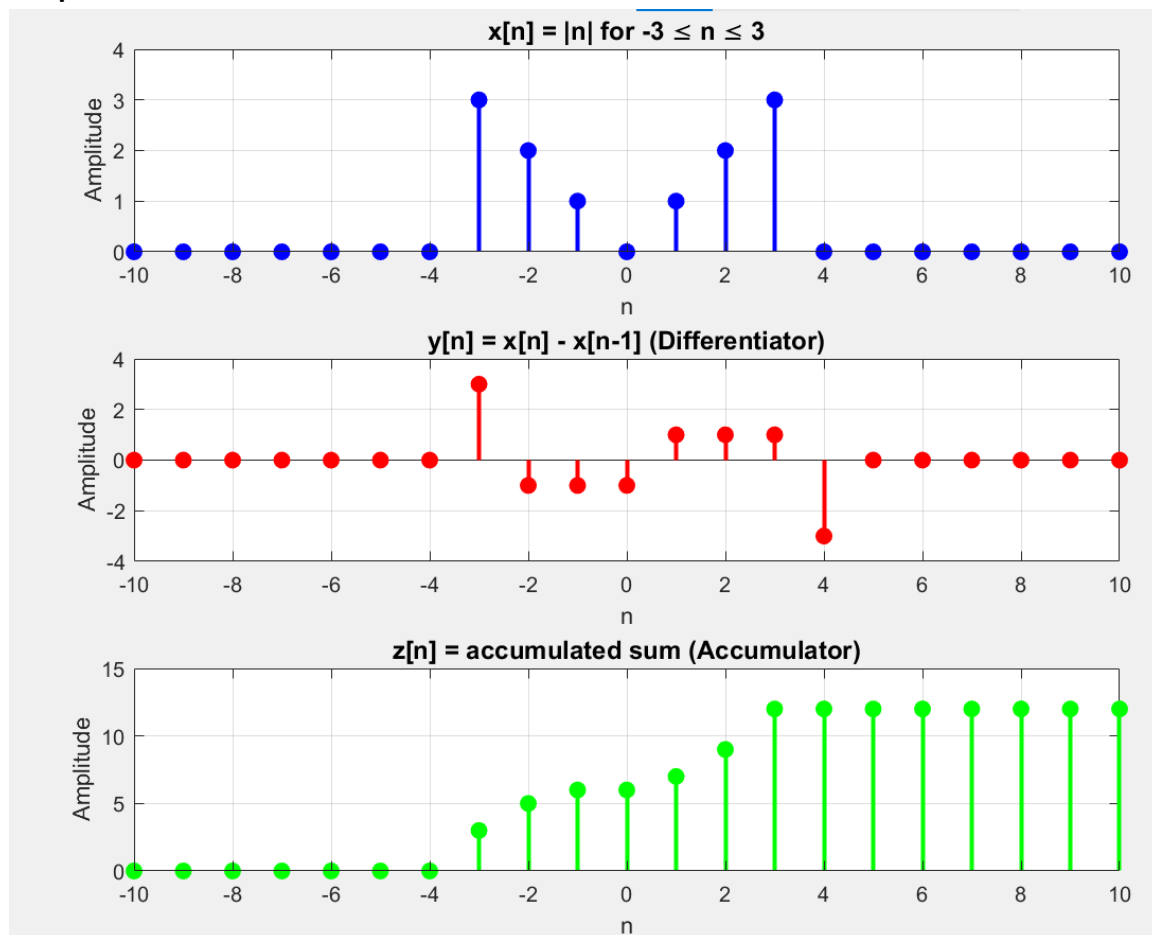
```

```

grid on;
ylim([-4 4]);
xlim([-10 10]);
% Plot z[n] (accumulator)
subplot(3,1,3);
stem(n, z, 'g', 'filled', 'LineWidth', 2);
title('z[n] = accumulated sum (Accumulator)', 'FontSize', 12);
xlabel('n'); ylabel('Amplitude');
grid on;
ylim([0 15]);
xlim([-10 10]);

```

Output:



Conclusion:

In this lab, the differentiator and accumulator systems were successfully implemented and analyzed for the discrete-time signal $x[n]=|n|$ defined over $-3 \leq n \leq 3$. Key findings include:

- The differentiator output $y[n]$ correctly identified the slopes of $x[n]$:
 - Positive slope for $n > 0$, giving $y[n]=1$
 - Negative slope for $n < 0$, giving $y[n]=-1$
 - Zero slope at $n=0$, where $y[n]=0$
- The accumulator output $z[n]$ effectively computed the cumulative sum, increasing symmetrically from $n=-3$ to $n=3$, with the maximum value occurring at $n=3$.